<u>NFS (Network File System) Server</u> <u>TASK – 8</u>

1. **Install the nfs-kernel-server package on your Linux VM and create a shared directory named /srv/music_share for sharing music files.**

   Ans :

   - # Update package index
     sudo apt update

   - # Install the NFS server package
     sudo apt install -y nfs-kernel-server

   - # Create the shared directory
     sudo mkdir -p /srv/music_share

   - # Set appropriate permissions
     sudo chmod 777 /srv/music_share

2. **Configure /etc/exports to share /srv/music_share with read and write permissions to a specific client IP (use your own second VM or localhost for testing), then restart the NFS service.**

   Ans :

   - # Install NFS server
     sudo apt update
     sudo apt install -y nfs-kernel-server

   - # Create the shared directory
     sudo mkdir -p /srv/music_share
     sudo chmod 777 /srv/music_share

- # Configure NFS export (replace 192.168.1.20 with your client VM IP or use 127.0.0.1 for localhost)
  echo "/srv/music_share 192.168.1.20(rw,sync,no_subtree_check)" | sudo tee /etc/exports

- # Apply the export configuration
  sudo exportfs -ra

- # Restart and enable the NFS service
  sudo systemctl restart nfs-kernel-server
  sudo systemctl enable nfs-kernel-server

- # Verify the export
  sudo exportfs -v

- # Verify the NFS service is running
  sudo systemctl status nfs-kernel-server

3. **On a client machine, mount the NFS share from your server to a local directory named ~/mnt/playlist and verify you can create and access files in the shared folder.**

   Ans :

   - # Install NFS client utilities
     sudo apt update
     sudo apt install -y nfs-common

   - # Create the mount point
     mkdir -p ~/mnt/playlist

   - # Mount the NFS share
     sudo mount 192.168.1.10:/srv/music_share ~/mnt/playlist

- # Verify the mount
  df -h | grep playlist
  mount | grep playlist

- # Create a test file
  touch ~/mnt/playlist/test_song.txt

- # Write data to the file
  echo "NFS share is working!" > ~/mnt/playlist/test_song.txt

- # Verify the file
  cat ~/mnt/playlist/test_song.txt

- # List the contents
  ls -l ~/mnt/playlist

4. **Simulate a Zomato-style shared menu folder: add a subfolder called 'restaurant_menus' inside your NFS share, copy any 2 PDF or image files into it, and confirm they are visible from the client mount.**

   Ans :

   On the NFS Server

   - # Create the subfolder inside the shared directory
     mkdir -p /srv/music_share/restaurant_menus

   - # Copy any 2 PDF or image files (replace with actual file paths)
     cp ~/Downloads/menu1.pdf
     /srv/music_share/restaurant_menus/
     cp ~/Downloads/menu2.jpg
     /srv/music_share/restaurant_menus/

- # Verify the files
  ls -l /srv/music_share/restaurant_menus

On the Client Machine

- # Confirm the folder is visible through the mounted NFS share
  ls -l ~/mnt/playlist/restaurant_menus

- # Open or view the files
  cat ~/mnt/playlist/restaurant_menus/menu1.pdf   # Optional (binary output)
  file ~/mnt/playlist/restaurant_menus/menu1.pdf
  file ~/mnt/playlist/restaurant_menus/menu2.jpg

5. **Un-mount the NFS share from the client and update your /etc/exports file to restrict access to read-only, then remount and test that you cannot create or modify files in the shared folder.

Hint: Use the 'ro' option in /etc/exports for read-only access.**

   Ans :

   On the Client Machine

   - # Unmount the NFS share
     sudo umount ~/mnt/playlist

   On the NFS Server

   - # Update /etc/exports to make the share read-on
     echo "/srv/music_share 192.168.1.20(ro,sync,no_subtree_check)" | sudo tee /etc/exports

- # Apply the changes
  sudo exportfs -ra

- # Restart the NFS service
  sudo systemctl restart nfs-kernel-server

- # Verify the export
  sudo exportfs -v

On the Client Machine

- # Remount the NFS share
  sudo mount 192.168.1.10:/srv/music_share ~/mnt/playlist

- # Verify the mount
  mount | grep playlist

- # Attempt to create a file (should fail)
  touch ~/mnt/playlist/read_only_test.txt

- # Attempt to modify an existing file (should fail)
  echo "Testing write access" >> ~/mnt/playlist/test_song.txt